

HYPERLOCAL JOB MARKETPLACE FOR TEMPORARY AND DAILY WAGE EMPLOYMENT

 **Foco Jobs**
"The Spark of a Single Shift, The Fire of Worker's Independence."

Find Jobs Log In Sign Up മലയാളം

"The Spark of a Single Shift, The Fire of Worker's Independence."

Welcome to PartWork Kerala

Find Jobs → Sign Up

500+
Jobs Posted

1200+
Job Seekers

7
Kerala Cities

98%
Verified Users

Why Foco Jobs?



Hyperlocal Matching

Find jobs within kilometers of your exact location — not just city level



Malayalam Support

Full Malayalam language interface for local workers across Kerala



Skill Gap Feedback

Get tips on exactly which skills you need to improve your chances



Trust Ratings

Verified bidirectional ratings between seekers and employers

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OUTLINE

- Introduction
- Problem & Objective
- Proposed Solution Overview
- Tech Stack and associated Software
- System Architecture / Design
- Implementation Details(Key Features and Demo)
- Evaluation & Analysis(with existing project)
- Conclusion
- Future Scope

INTRODUCTION

What is Foco Jobs?

- A full-stack MERN web application serving as a hyperlocal part-time job platform connecting job seekers and employers across 7 Kerala cities.

Why is it needed?

- Existing platforms like Internshala and Apna are English-only, use city-level filtering, and provide no feedback or trust system

PROBLEM AND OBJECTIVE

Problem Statement

- Existing platforms use city-level filtering only, are English-only, provide no feedback to rejected applicants, and have no trust or verification system for part-time workers.

Objectives

- Build GPS-based proximity job search
- Skill-gap feedback on applications
- Bidirectional trust ratings
- OTP email verification for seekers

PROPOSED SOLUTION OVERVIEW

Geospatial Proximity

- MongoDB 2dsphere index with \$near query returns jobs sorted by real GPS distance, not city name.

Malayalam Bilingual UI

- react-i18next with en.json and ml.json. One-click language toggle, persisted in localStorage.

Skill-Gap Feedback

- TF-IDF keyword match scores the application and shows missing skills to the seeker.

Trust Rating

- Bidirectional 1–5 star rating after job completion. Average recalculated and shown on all profiles.

OTP Verification

- 6-digit OTP via Gmail SMTP. Seekers cannot apply until email is verified.

TECH STACK AND ASSOCIATED SOFTWARE

Frontend

- React.js — SPA with React Router, Axios, AuthContext (JWT state), react-i18next (bilingual)

Backend

- Node.js + Express.js — REST API on port 5000, JWT middleware, bcrypt, CORS, Nodemailer

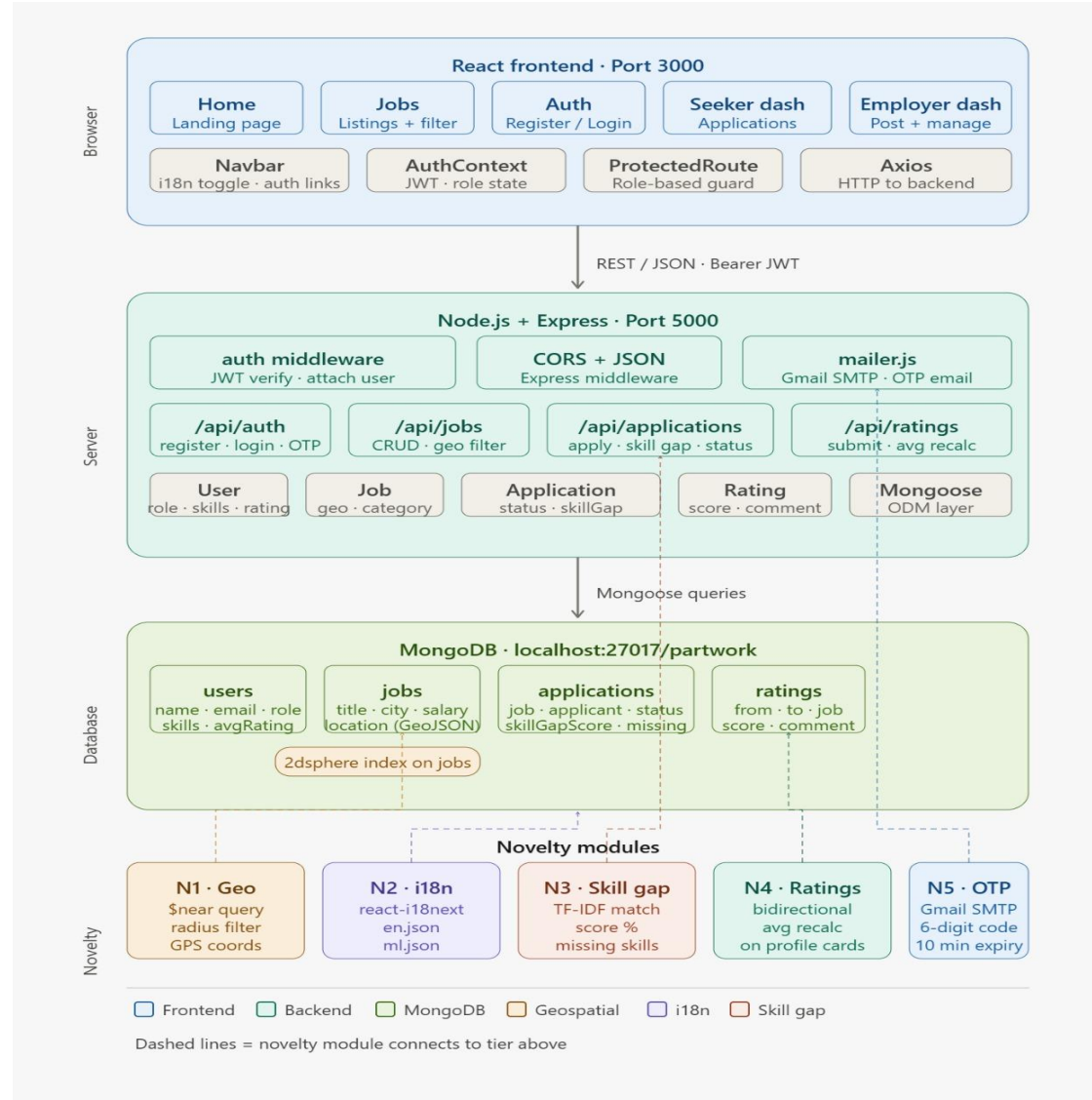
Database

- MongoDB — NoSQL with Mongoose ODM, 2dsphere geospatial index on jobs collection

Security & Auth

- JWT — stateless auth tokens • bcrypt — password hashing • OTP — email verification

SYSTEM ARCHITECTURE



IMPLEMENTATION DETAILS

Geospatial Search

- Employer GPS stored as GeoJSON Point with 2dsphere index.
- \$near + \$maxDistance returns jobs sorted by real distance.

Skill-Gap Engine

- Tokenize job description and seeker skills → remove stopwords → set intersection → $\text{score} = \text{matched}/\text{total} \times 100$ → stored in Application document.

OTP Verification

- 6-digit OTP stored with 10-min expiry on register.
- Gmail SMTP via Nodemailer. Seekers blocked from applying until verified.

Trust Rating

- After job complete, both parties rate 1–5 stars.
- Average recalculated using reduce() and updated on user profile.

APPLICATION SCREENSHOTS

Welcome back
Mathew Shaji 🍌
Verify your email to apply for jobs

Find Jobs →

2
Total Applied

0
Pending

0
Accepted

1
Completed

My Applications

Driver Required

Ajmal · Kollam
₹5000

Rejected
5/3/2026

Skill Gap Score: 0% match
Missing Skills: drive, everyday, morning, 3hrs, travel

Retail sales

Lulu Mall · Thiruvananthapuram
₹12000

completed
4/25/2026

Skill Gap Score: 0% match
Missing Skills: need, retail, sales, associate, customer

Employer Dashboard
Kevin Jose 📄
3

+ Post a Job

3
Active Jobs

0
Total Applicants

3
Rating

Your Job Postings

driver

Kollam · Delivery · ₹12000

View Applicants

Retail sales

Thiruvananthapuram · Retail · ₹12000

View Applicants

: Retail Sales Associate

Kochi · Retail · ₹12000

View Applicants

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EVALUATION AND ANALYSIS

Results Achieved

- Role-based dashboards
- GPS proximity search working
- Skill-gap scores displayed
- Bidirectional ratings updating in real time

Comparison with Existing Platforms

Feature	Foco Jobs	Internshala	Apna	QuikrJobs
GPS Proximity	✓	X (City-only)	X (City-only)	X (City-only)
Malayalam UI	✓	X	X	X
Skill-gap Feedback	✓	X	X	X

CHALLENGES AND SOLUTIONS

Challenge 1: Geospatial Query Returning Empty Results

- The \$near query returned no results even though jobs existed. Resolved by correcting coordinate order and adding jobSchema.index({ location: '2dsphere' }).

Challenge 2: JWT Token Lost on Page Refresh

- Users were being logged out on every page refresh because the token was stored only in React state. Resolved by persisting the token in localStorage and restoring it in AuthContext using useEffect on app startup.

Challenge 3: Malayalam UI Not Switching Instantly

- Some components were not re-rendering in Malayalam after language toggle. Resolved by replacing all hardcoded text with t('key') function calls across all components.

CONCLUSION

- Foco Jobs is a fully functional MERN platform built for Kerala's informal part-time workforce with five distinct novel contributions not found combined in any existing Indian job platform.

What was achieved

- GPS proximity search • Malayalam bilingual UI • Skill-gap feedback • Bidirectional trust ratings • OTP email verification — all working end to end.

Significance

- First hyperlocal, bilingual, skill-aware, trust-verified part-time job platform designed specifically for Kerala's informal labor market.

FUTURE SCOPE

Real ML Skill Matching

- Replace TF-IDF with HuggingFace sentence-transformers (MiniLM) for semantic similarity.

Shift-Slot Scheduling

- Employers post time slots, seekers mark availability, system auto-matches overlap.

Mobile App

- React Native cross-platform app for Android and iOS.

In-App Messaging

- Real-time chat via Socket.io between employers and applicants.

THANK YOU